## UME718 : Applied Thermodynamics

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| 3 | 1 | 2* | 4.0 |

**Course Objective**: This course introduces various thermodynamic cycles used for analysis of heat engines and gas turbines. This course also introduces fundamental thermodynamic operating principles, phenomena of I.C. engines and performance parameters of I.C. engines.

### Syllabus

**Thermodynamic Air Standard Cycles:** Introduction, Assumptions, Carnot cycle, Stirling cycle, Ericsson cycle, Otto cycle, Diesel cycle and Dual cycle. Comparison of Otto, Diesel and Dual cycles.

**Brayton cycle:** Open and closed cycles, Effect of pressure ratio, Brayton cycle with inter-cooling and reheating, Brayton cycle with regeneration, Applications of Brayton cycle, Comparison between Otto and Brayton cycles

**I.C.Engines:** Introduction, classification and application, Combustion in S.I. engine: Flame propagation, pre-ignition, detonation, engine variables effects, mixture requirements, fuel rating; Fuel supply system, Combustion in C.I.Engine, delay period, knocking, engine variables effects, fuel requirements, rating, combustion chambers; Fuel supply system, Engine cooling and lubrication, Performance of engines: Variable and constant speed tests as per ISI standards. Performance curves, heat balance, Exhaust and emission control.

**Steam generators and condensers:** Introduction to steam generator, mountings and accessories, condensers, Overview of steam power plant, boiler trial -energy balance sheet.

### Laboratory Work

Assembly of Petrol and Diesel engine components, Identification of components and their working principles, Study of performance of Petrol and Diesel engines (Kirloskar Diesel engine, Ruston Diesel engine, Krimo oil engine, VCR engine, Dual fuel engine)

**Micro Project:** Students in group of 4/5 will carry out micro project on alternative fuels to be used in I.C. engines (Petrol, Diesel and Dual fuel engines)

### Course Learning Objectives (CLO)

The students will be able to:

1. Derive and analyze Otto cycle, Diesel cycle and Dual cycle air standard efficiencies.
2. Derive and analyze simple Brayton cycle.
3. Determine and analyze the performance parameters of I.C. engines in an engine test rig.
4. To prepare heat balance sheet of the boiler.

**Text Books**

1. Pulkrabek, W. W., Engineering Fundamentals of Internal Combustion Engines, Pearson education Asia, New Delhi (2007).
2. Vasandani, V. P. and Kumar, D. S., Heat Engineering, Metropolitan Book Company, New Delhi (2003).

**Reference Books**

1. Heywoold J. B., I.C. Engine, McGraw Hill, New Delhi (1988).
2. Joel, R., Basic Engineering Thermodynamics, Pearson Education Asia, New Delhi (1996).
3. Granet, I., Thermodynamics & Heat Power, Pearson Education Asia, New Delhi (2003).
4. Ganeshan, V., Internal Combustion Engines, Tata McGraw Hill, New Delhi (2007).
5. Nag, P. K., Power Plant Engineering, Tata McGraw Hill, New Delhi (2008).

**Evaluation Scheme**

| S. No. | Evaluation Elements | Weightage (%) |
|---|---|---|
| 1. | MST | 25 |
| 2. | EST | 45 |
| 3. | Sessionals (Assignments/ Projects/ Tutorials/Quizzes/Lab Evaluations) | 30 |